



Towards Automation of Reference Data Generation for ADAS/AD Functions Development – ALiVA Framework

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Abstract

The advancements towards autonomous driving have propelled the need for reference/ground truth data for development and validation of various functionalities. Traditional data labelling methods are time consuming, skills intensive and have many drawbacks. These challenges are addressed through ALiVA (automatic lidar, image & video annotator), a semi-automated framework assisting for event detection and generation of reference data through annotation/labelling of video & point-cloud data. ALiVA is capable of processing large volumes of camera & lidar sensor data.

Main pillars of framework are object detection-classification models, object tracking algorithms, cognitive algorithms and annotation results review functionality. Automatic object detection functionality creates a precise bounding box around the area of interest and assigns class labels to annotated objects. Object tracking algorithms tracks detected objects in video frames, provides a unique object id for each object and performs distance ranging.

A unique feature of cognitive algorithms is the elimination of non-realistic objects of interests which appear in billboards or advertisements on buses/trucks.

The framework also has a feature of event detection like overtaking scenarios or pedestrians/animals crossing the roads.

Annotation review functionality is provided where assessment and correction of auto annotated data can be done manually. The results can be saved in standard file formats such as txt, csv, Json and open ASAM, ensuring compatibility across different systems.

ALiVA replaces traditional annotation methods, thereby reducing the effort, the need for skilled resources and the time required to annotate large datasets. This eliminates human biases, manual errors and inconsistencies.

ALiVA is validated for numerous customer requirements and offers a large amount and variety of data to quantify the benefits offered. Some of the distinguishing features are models and functionalities that are optimized for Asian road scenarios, which are typically characterized by very high road density. It is platform independent, adaptable to newer requirements, complements newer event definitions for data segmentation and works both in cloud environments for Data as a service and as a standalone desktop application.

Introduction

Automated driving (AD) and Advanced Driver Assistance Systems (ADAS) are reshaping the automotive industry, transcending traditional mechanical and electrical engineering boundaries[1]. This transformation extends into diverse fields, including computer science, information technology (IT), and artificial intelligence (AI). AD encompasses two critical aspects: (1) computerized driving, which involves controlling vehicle functions, and (2) navigation, which encompasses motion planning and environmental awareness. ADAS components, such as acceleration and braking control, lane departure warnings, cruise control, play pivotal roles in achieving these objectives. Leveraging data from sensors like Light Detection and Ranging (LIDAR) and cameras, ADAS systems continuously assess the vehicle's state and its surroundings.

To manage the complexity of real road scenarios, ADAS algorithms rely on high quality artificial intelligence models and machine learning. The performance of ADAS functionalities depends on these models, likewise performance of these AI models is inherently restricted by the quality and quantity of available training and validation data (ground truth data). A key bottleneck in achieving improvements in any application domain is the availability of ground truth data, with sufficient reliability, complexity, and coverage.

There are various traditional data labelling methods available for creating annotations that can be considered as ground truth data[2]. However, these traditional methods involve Human-in-the-Loop (HiL) approach i.e. manually annotating video/image frames and 3D point cloud data. Vision and point-cloud data annotation is an intense task due to the amount of information they contain.

In addition to the large amount of information contained in a single image, with video we must consider the temporal component, changes in the perspective of the scene, and the relationships of objects between images. Therefore, for every hour of collected visual point cloud data, more than 500 hours of human intervention are required to annotate the images and point cloud data separately.

There is a wide variety of annotation tools that boost HiL, with different characteristics depending on the functionality they provide and how they operate. Among the most common annotations, available in almost all tools, for example is the ability to easily draw a rectangle or an ellipse. Despite the extensive diversity of the annotation tools available, there is still a need for new custom semi-automatic annotation framework that meets the computational complexity and the data scale to efficiently address vision and point-cloud annotation for generating ground-truth data. In this scenario, contributions of this semi-automated framework for ground truth data generation can be summarized as follows:

- An innovative architecture to integrate annotation workflow as a stand-alone application providing data as a service (DaaS).
- Efficient processing of large volumes of sensor data, including images, LiDAR point clouds.
- Rapid, reliable and automatic ground truth data generation for various on road complex scenarios (especially Indian roads) using indigenous in house developed AI models and cognitive algorithms.
- Rich annotation reviewing functionality to manually examine, edit, correct & delete annotations.
- Up to 80% reduction in annotation efforts and time while using ALiVA framework compared to traditional methods, to generate the ground truth data in complex AD/ADAS annotation pipelines.
- Built-in object tracking, cognitive filtering, and event/scenario detection functionalities making ALiVA applicable across different levels of AD/ADAS.

This paper is structured to provide a comprehensive understanding of the framework, its functionalities, and its validation across numerous client requirements. First, the technical architecture and main components of the framework are explained and an insight into the underlying algorithms and methods used is given. It then delves deeper into the algorithmic intricacies of object detection, tracking, and event detection, and explains how each of these areas contributes to the overall goal of efficient data annotation. Furthermore, ALiVA's adaptability to diverse road scenarios, particularly in Asian settings characterized by high road density and its use as Data as a Service (DaaS) is highlighted. Lastly, the framework's deployment options, including its compatibility with cloud environments and standalone desktop applications are discussed.

The importance of the framework in revolutionizing data annotation methods for autonomous driving applications is emphasized. A careful literature review and the

formulation of goals and motivations are the foundation for the investigation of framework capabilities described in this paper.

Reference Data Generation and Its Significance

In the context of Advanced Driver Assistance Systems (ADAS), reference data includes a variety of sensory inputs, including image and LiDAR (Light Detection and Ranging) data. These diverse datasets play a fundamental role in the development and validation of ADAS functionalities, offering insights into the surrounding environment and enabling the creation of robust perception algorithms. Images provide detailed visual information about the road scene, vehicles, pedestrians, and signage, facilitating tasks such as object detection, recognition, and classification. Video sequences capture temporal dynamics and contextual information, essential for tasks like tracking, motion estimation, and scene understanding. It can also be used to identify events and scenarios and to generate metadata. LiDAR data, on the other hand, offer precise 3D spatial information about the surrounding objects and terrain, enabling accurate depth perception, obstacle detection, and localization. By integrating these complementary data modalities, reference datasets empower researchers and engineers to simulate diverse driving scenarios, train machine learning models, and to validate ADAS systems across a spectrum of real-world conditions, ultimately advancing the safety and reliability of autonomous vehicles.

Data Preprocessing

Data preprocessing for raw sensor data involves several key steps. Firstly, data segregation is crucial to handle continuous streams efficiently, where data is segmented into manageable chunks or frames based on time intervals or specific events. This ensures that each segment represents a distinct scenario for annotation, leading to a more relevant dataset. Secondly, ensuring data variety is essential, covering different scenarios like lighting and weather conditions, traffic densities, object interactions, road types, infrastructure, pedestrian, animal scenarios, vehicle maneuvers, and object occlusions[2]. This variety helps train annotation models to generalize well to real-world situations. Thirdly, synchronizing lidar and camera data is necessary when captured separately to align them accurately, typically achieved using timestamp information. Additionally, data refinement techniques are applied to eliminate noise, outliers, and artifacts, which could reduce annotation accuracy and impact AI model effectiveness for ADAS functionalities. Finally, efficient data storage solutions are required due to the high volume of data generated by lidar and camera sensors, encompassing short-term storage for preprocessing and annotation, as well as long-term storage for archiving annotated datasets.

Data Annotation

Data annotation is the process of detection, identification, and labeling objects in image or point cloud data. Annotations are often represented using bounding boxes or segmentation masks, outlining the spatial extent of each object. This meticulous labeling process is crucial for training AI models to recognize and understand objects in sensor data accurately. Lidar and vision data require different annotation approaches due to their distinct data formats. Lidar data is inherently three-dimensional, capturing spatial information about objects in the scene. Therefore, annotating lidar data involves labeling objects in 3D space, which requires annotators to define the object's position and dimensions accurately. On the other hand, camera data captures images, requiring annotations to be made directly on the image plane. This means that objects in camera data are annotated using 2D bounding boxes or segmentation masks, which provide a simpler representation of the object's location and shape. In traditional annotation-labelling process human annotators are responsible for identifying and labeling objects of interest manually within sensor data.

Traditional Annotation Methodology Challenges and Need of Automation

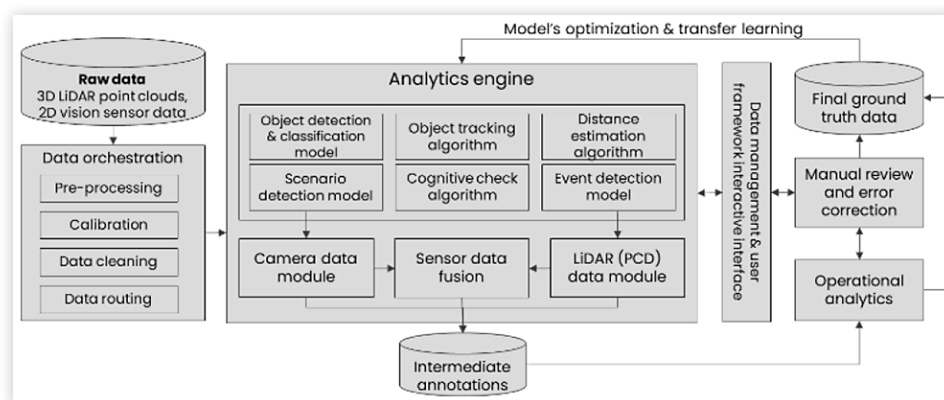
The conventional manual data labeling method poses several challenges and pain points. Annotators may interpret labeling criteria differently, leading to inconsistencies. Additionally, manual annotation is time-consuming and costly, particularly for large datasets, and can result in delays. Handling object class imbalances, especially for rare classes, can be challenging. Annotating objects with temporal or spatial continuity requires expertise, as does annotating dynamic environments with moving objects. Data annotation for machine learning introduces complexities such as feature selection and class imbalance

handling. Privacy and security concerns must also be addressed, as annotated datasets may contain sensitive information. Tool limitations can impact annotation efficiency and quality, and maintaining annotation consistency over time can be difficult. Quality control measures to ensure annotation quality are resource-intensive and may require additional expertise. The need for automated annotation tools arises from the challenges posed by manual annotation methods[3]. Automated tools can significantly improve the efficiency and accuracy of the annotation process. They can interpret labeling criteria consistently, reducing inconsistencies among annotators. Automation can also reduce the time and cost associated with manual annotation, especially for large datasets, and can help address object class imbalances by efficiently annotating rare classes with various annotation types[4]. Automated tools can handle annotations requiring temporal or spatial continuity and can effectively annotate dynamic environments with moving objects. Additionally, automation can streamline the data annotation process for machine learning, addressing complexities such as feature selection and class imbalance handling. Automated tools can also enhance data privacy and security by providing mechanisms for anonymization and secure data handling. Overall, automated annotation tools can overcome the challenges and pain points of manual data labeling, making the annotation process more efficient, accurate, and cost-effective.

ALiVA (Automated Lidar, Image, Video Annotator) Framework – System Architecture

ALiVA framework is designed to integrate heterogeneous annotation steps as a stand-alone application. Figure 1 illustrates a highly scalable & reliable system architecture is developed.

FIGURE 1 System architecture diagram



Data Orchestration

In ALiVA data orchestration plays a pivotal role in managing data flow across various processing stages. These stages include data pre-processing, calibration, cleaning, and routing, all of which contribute to the quality and accuracy of annotated data. Raw sensor data from cameras, LiDAR, and other sensors undergoes pre-processing. Tasks like noise reduction, normalization, and alignment ensure consistency across different sensor modalities. Accurate alignment and synchronization of data from different sensors are crucial. Calibration corrects sensor biases, time delays, and other factors affecting data accuracy. Removing errors and irrelevant information ensures high-quality annotated data. Clean data is essential for optimal ADAS algorithm performance. After pre-processing, calibration, and cleaning, data is routed to relevant modules for annotation. Algorithms direct data flow, ensuring correct and efficient annotation.

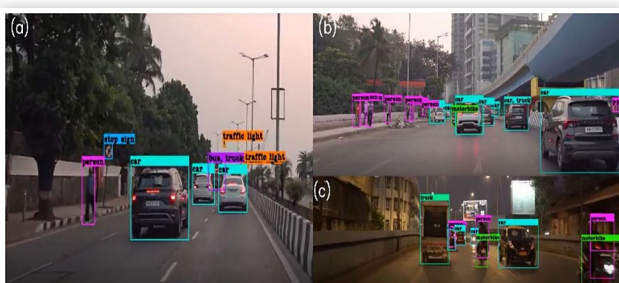
Auto Annotation Analytics Engine

Image Data Annotation

As shown in Figure 2 this component is designed to automatically annotate the objects in video frames and images. The input video is mp4 and is converted into frames based on the frame rate of the input video, eliminating the risk of information loss.

This module uses object detection model that utilizes convolutional neural networks (CNNs) with a cross-stage partial network (CSPNet) backbone. This model architecture enhances CNN's learning capability while reducing computational complexity. Additionally, a neck structure is introduced to combine features and perform feature fusion, improving the model's ability to detect objects at various scales. The object detection head predicts bounding boxes, class probabilities, and confidence scores. Post-prediction, non-maximum suppression (NMS) is applied to eliminate duplicate detections. Annotated objects are assigned unique object id & are tracked in

FIGURE 2 Illustrative image annotations – (a) Evening scenarios (b) Day scenarios (c) Night scenarios



consecutive video frames, resulting in increased accuracy. In addition, a tracking algorithm is used to identify objects across multiple frames to increase accuracy.

The model is trained on a combination of open-source datasets, implementing a multi-modal approach covering multiple ADAS-related classes such as cars, bikes, trucks, buses, pedestrians, autorickshaws, traffic lights, and traffic signs etc[5, 6, 7].

LiDAR Data Annotation

This module is specifically tailored for annotating 3D point-cloud data as referred in Figure 3. Like the other module, it features an auto-annotation functionality that utilizes AI-based object detection-classification model. Additionally, it provides visualization capabilities for point clouds and incorporates sensor fusion functionality to visualize and annotate both vision (frames) and lidar (pcd) data.

The model takes a point cloud as input, which is essentially a collection of points in 3D space represented by their (x, y, z) coordinates. This point cloud undergoes processing by a 3D backbone network, which comprises 3D convolutional neural networks. The purpose of this backbone network is to encode and transform the raw point cloud into a higher-level representation suitable for feature extraction. Additionally, a 'neck' component is introduced that processes the features extracted by the backbone network. This neck component performs feature fusion and refinement operations, enhancing the model's ability to detect objects.

The resulting feature representation is then fed into the object detection head, which consists of a series of convolutional layers. The object detection head generates anchors (3D bounding boxes) based on the features. Subsequently, the model predicts offsets and dimensions for these anchors, ultimately producing the final bounding box predictions. During training, optimization of the model is done using a loss function that combines regression losses (for bounding box regression) and classification losses (for objects and class prediction)[8, 9].

Sensor Fusion

In the annotation framework, the fusion process involves aligning and synchronizing the data streams from both sensors, followed by sophisticated algorithms that merge

FIGURE 3 Lidar data annotation

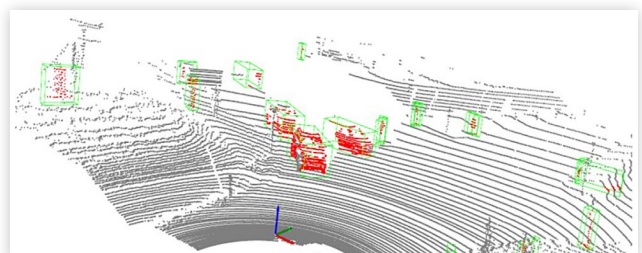
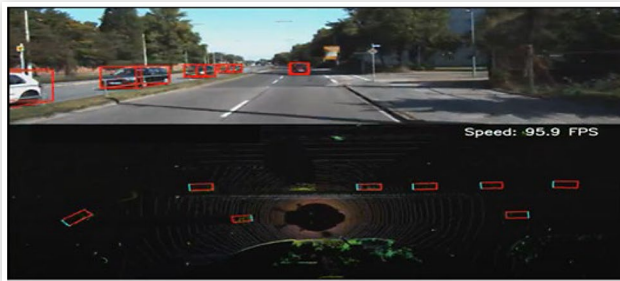


FIGURE 4 Camera & LiDAR sensor fusion

and interpret the data. This fusion enhances the overall perception capabilities of the framework, enabling it to accurately annotate various driving scenarios.

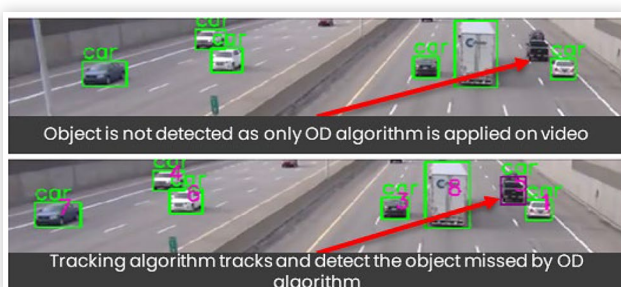
Figure 4 shows integration/fusion of camera and LiDAR sensor data which is crucial for enhancing the annotation process. Camera sensors capture high-resolution images, providing detailed visual information essential for object recognition and classification. On the other hand, LiDAR sensors offer precise 3D spatial data, aiding in the accurate localization of objects in the environment.

Models and Algorithms

Object Tracking

Object tracking algorithm serves as a critical differentiator in ALIVA. The framework is leveraging second equation of motion and Kalman filter technique to track objects across frames in a sequence of images [10]. It is particularly effective in scenarios where there is uncertainty in the state of the object being tracked and where measurements of the object's state are noisy.

As shown in Figure 5 this algorithm initializes tracking by selecting detected objects in first frame of the video sequence. Unique object id is assigned to each object. The Kalman Filter represents the state of a tracked object by using a state vector x and a state transition matrix A . This vector typically includes the object's position and velocity. In the prediction step, given the current state estimate and the state transition matrix A , the Kalman

FIGURE 5 Object tracking for accurate and consistent detection

Filter predicts the object's state at the next time step based on its motion dynamics (e.g., constant velocity or acceleration). The measurement update step incorporates new measurements z of the object's state, adjusting the estimate. The Kalman gain is calculated to balance the prediction and measurement information. Finally, the state update and error covariance update steps refine the estimate iteratively.

In multi-object tracking scenarios, data association techniques based on appearance and motion cues are used resulting in algorithm associating objects across frames to maintain consistent tracks. The algorithm handles occlusions, where the object is temporarily hidden by other objects, by predicting the object's position based on its motion model and appearance model. Deletions, where the object disappears from the scene, are also handled by predicting the object's position until it reappears.

Multi Modal Approach

AI democratization is a cutting-edge frontier in the field of artificial intelligence. Multi-model deep learning architectures play a pivotal role in this domain, as they can handle diverse types of input data. Each model possesses its own unique intelligence, and when these intelligences collaborate, they lead to customization. The framework leverages a network of models, effectively democratizing intelligence for tasks such as animal object detection, traffic signal and sign recognition, and other significant objects. By executing these models sequentially and merging their annotations into a unified CSV, JSON and ASAM formats, we enhance the accuracy of object annotations in both images and video frames.

Events Detection

The implementation of event detection within the ALIVA framework entails a systematic approach to identify and analyze significant occurrences in visual data streams. Leveraging advanced machine learning algorithms and computer vision techniques, it also detects events such as traffic sign and signal occurrences, pedestrian activities (e.g., crossing roads or walking on footpaths), and the presence of animals on roads. Additionally, the framework is specifically tuned for Indian roads, considering unique challenges and characteristics of Indian traffic environments.

Scenario Identification/Detection

One of the key components of the semi-automated annotation framework for ADAS is the ability to detect and identify various driving scenarios. These scenarios play a crucial role in determining the appropriate actions and responses of the ADAS system. The framework incorporates advanced algorithms and sensor fusion techniques to accurately identify various scenarios such as rainy

condition, foggy conditions, wet road surface, road surfacing condition, visibility assessment. The framework's ability to accurately detect and identify these scenarios enhances the quality of ground truth data.

Cognitive Check Algorithm

One of the distinctive features provided by cognitive algorithms lies in their ability to filter out non-realistic objects of interest that might appear in billboards or advertisements on buses and trucks. By leveraging advanced image processing techniques such as image filtering & masking, these algorithms enhance the accuracy of object detection, ensuring that only relevant and contextually appropriate elements are identified and considered for further processing. This capability contributes significantly to improving the overall quality and reliability of visual recognition systems, making them more robust in real-world scenarios.

Distance Estimation

Illustrated in Figure 6, distance estimation algorithm relies on monocular depth estimation which involves predicting the depth map of a scene using a single camera. Deep learning models are commonly used for this purpose. Once the depth map is generated, the algorithm calculates the distance to objects in the scene based on the depth values. This process includes camera calibration and geometric transformations to convert pixel distances into real-world distances. This application does not require distance to be measured with extremely high accuracy. Thus, for the purpose of safety distance monitoring with some deviations in fraction of a meter are sufficient[11].

Annotation Result Review

Output generated by auto-annotation process are typically to be reviewed by experts for quality check. This manual review functionality serves as a critical checkpoint, ensuring the accuracy and relevance of annotations generated through automated processes. Such feature embedded in ALiVA framework empowers human

FIGURE 6 Real time distance estimation with monocular vision

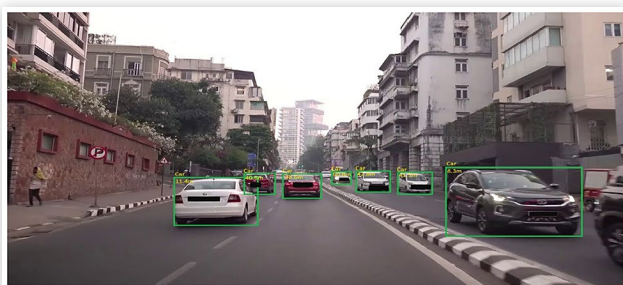
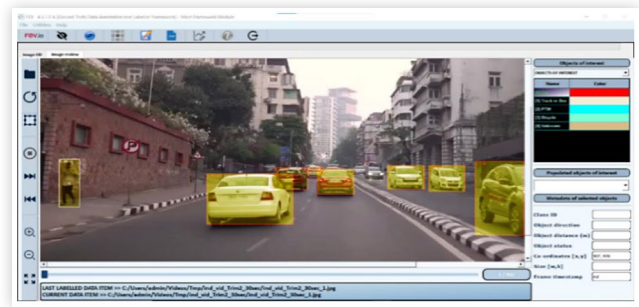


FIGURE 7 Expert manual review functionality



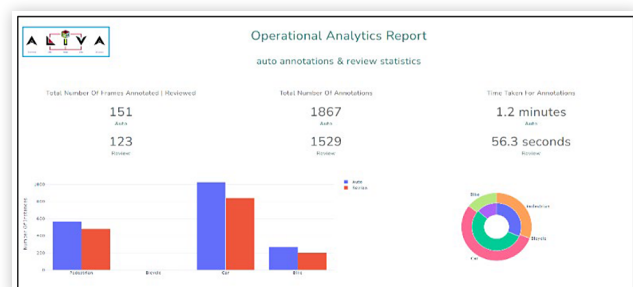
annotators to meticulously inspect and validate class labels assigned to objects, while also allowing for adjustments such as boundary refinement, resizing bounding boxes, and label modification. Moreover, annotators have the flexibility to supplement automated annotations by manually adding missing annotations, deleting misclassified objects, and editing/adding meta data, thereby enriching the completeness and quality of the datasets as shown in Figure 7. Tools in framework for these review operation boost the overall performance of the process.

One of the differentiating feature of the review functionality is tracking of review actions, generating process analytics reports based on the same, and also routing the corrective information for retraining/tuning the models and algorithms automatically – a self-learning architecture.

Operational Analytics Report

The operational analytics functionality in the annotation framework enables a comprehensive comparison between annotation done in auto annotation process and manual correction based on various key metrics as illustrated in Figure 8. This functionality allows users to evaluate the efficiency and effectiveness of each annotation step by comparing the time taken for annotation, the total number of annotations generated, the total number of classes identified, and the distribution of annotated objects across these classes. Concurrently, an operation analytics report is automatically generated, offering in-depth insights into various metrics essential for data annotation, including annotation distribution,

FIGURE 8 Illustrative operational analytics report



trend analysis, and the total duration required for both automated annotation and manual review. This comprehensive approach not only enhances the accuracy and completeness of annotations but also streamlines the annotation workflow, ensuring the efficacy of the annotation process.

Visualization

ALiVA integrates advanced visualization capabilities to offer users intuitive and insightful displays of annotated data. Leveraging cutting-edge techniques, the framework enhances the annotation process by providing clear and informative representations of annotated objects and their associated metadata. Color-coded bounding boxes visually differentiate different object categories, while object class labels provide descriptive information. Additionally, metadata display features offer contextual information alongside annotated objects, contributing to a comprehensive annotation experience.

Evolving Architecture

The amalgamation of automated and manual annotation culminates in the creation of ground truth data, representing a meticulously curated dataset that reflects accurate annotations verified by human expertise. This ground truth data serves as the benchmark for model optimization.

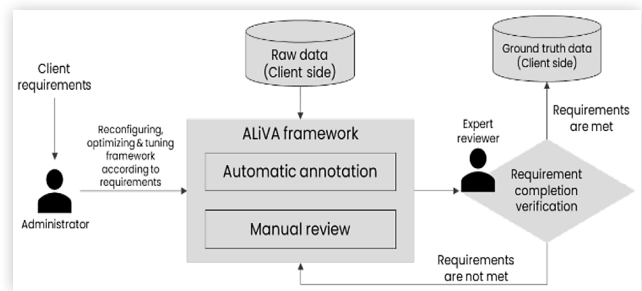
The iterative nature of the framework's architecture enables continuous improvement through transfer learning techniques. Leveraging the ground truth data, pre-existing models undergo retraining and optimization processes to adapt and refine their predictive capabilities. Transfer learning facilitates the seamless integration of knowledge acquired from previously annotated data, enabling models to enhance their accuracy and adaptability over time.

The evolving architecture of the framework encompasses mechanisms for feedback loops, allowing for the incorporation of new annotations and insights into subsequent iterations of model training. This adaptive approach ensures that the framework remains agile and responsive to evolving data patterns and requirements.

Data as a Service (DaaS)

The proposed framework for semi-automatic annotation is designed to be versatile and adaptable to different deployment scenarios as. It can function both as a standalone system and as a cloud-based service, offering flexibility in deployment options based on the specific needs and preferences of users as reflected in [Figure 9](#). When deployed as a standalone system, the framework can

FIGURE 9 Data as a service



be installed and operated within an organization's infrastructure, providing complete control over data management and annotation processes. This deployment option is suitable for users who prioritize data privacy and security, as all data processing and storage remain within their premises or private cloud environment. On the other hand, the framework can also be deployed as a cloud-based service, leveraging the scalability and accessibility advantages of cloud computing. In this deployment model, users can access the framework over the internet, allowing for easy collaboration and remote access to annotation tasks. Cloud deployment is particularly beneficial for users who require flexible computing resources and want to reduce infrastructure costs.

ALiVA Differentiators

As a derived benefits of digital transformation and evolving AI technology coupled with engineering analytics paradigm, ALiVA offers key differentiators over effortful and costly traditional methods. This is summarized in [Table 1](#).

Summary

The semi-automatic annotation framework presented in this paper offers a promising approach for accelerating the generation of ground truth data. By combining the strengths of automated algorithms and human annotators, the framework addresses the challenges of efficiency and accuracy in data annotation, which are critical for training robust ADAS models. Through the integration of state-of-the-art artificial learning algorithms for object detection, such as physics and mathematics-based approaches for object tracking, monocular depth estimation for distance estimation, and sensor fusion for comprehensive environment perception, the framework enables the generation of high-quality annotated datasets. The framework's semi-automatic nature not only enhances annotation efficiency but also ensures annotation quality by leveraging human annotators' expertise. The framework's versatility allows it to be applied to generate ground truth data for a wide range of ADAS

TABLE 1 ALiVA differentiators over traditional methodologies

Parameter	Traditional methods	ALiVA
Productivity	600 objects/hour	1,800,000 objects/hour
Effort	100%	10%
Addition of new Event/scenario	Complex and effortious	Simple with lesser efforts
Number of object types	Limited	Flexible (according to requirements)
Resource requirement	High (labor & skill intensive)	Low
Complex task (addition of meta data, KPI's) handling	Limited (simple task)	High (can handle complex task)
Label consistency	Low to Moderate, prone to obvious human errors	High
Adaptability to new object class/scenario	Low to Moderate (Need to train nnotators for new objects)	High (can accommodate new class/scenarios with some reconfiguration)
Annotation cost	High	Low to Moderate (Initial setup cost)
Quality control	Manual oversight required	Built-in mechanisms with expert review
Data privacy	Low	High
Scalability	Low	High (scalable for different annotation requirements)
Error handling	Manual intervention required	Automated error handling

functionalities and scenarios. Its ability to adapt to different sensor configurations, environmental conditions, and annotation requirements makes it well-suited for the complex and dynamic nature of real-world driving scenarios. By streamlining the annotation process and improving annotation quality, the framework has the potential to accelerate the deployment of safer and more advanced ADAS systems, ultimately contributing to the goal of reducing accidents and saving lives on the road.

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